

3 A student studied the water potential of carrot tissue.

Pieces of weighed carrot tissue were placed in a range of sucrose solutions.

After two hours, the tissue was blotted and weighed again. The results are shown in the table.

Concentration of sucrose solution /mol dm ⁻³	Original mass of carrot tissue /g	Final mass of carrot tissue /g	Percentage change in mass of carrot tissue (%)
0.0	2.40	3.10	29.17
0.2	2.48	2.68	8.06
0.4	2.48	2.42	-2.42
0.6	2.31	1.96	-15.15
0.8	2.41	1.78	-26.14
1.0	2.75	1.92	

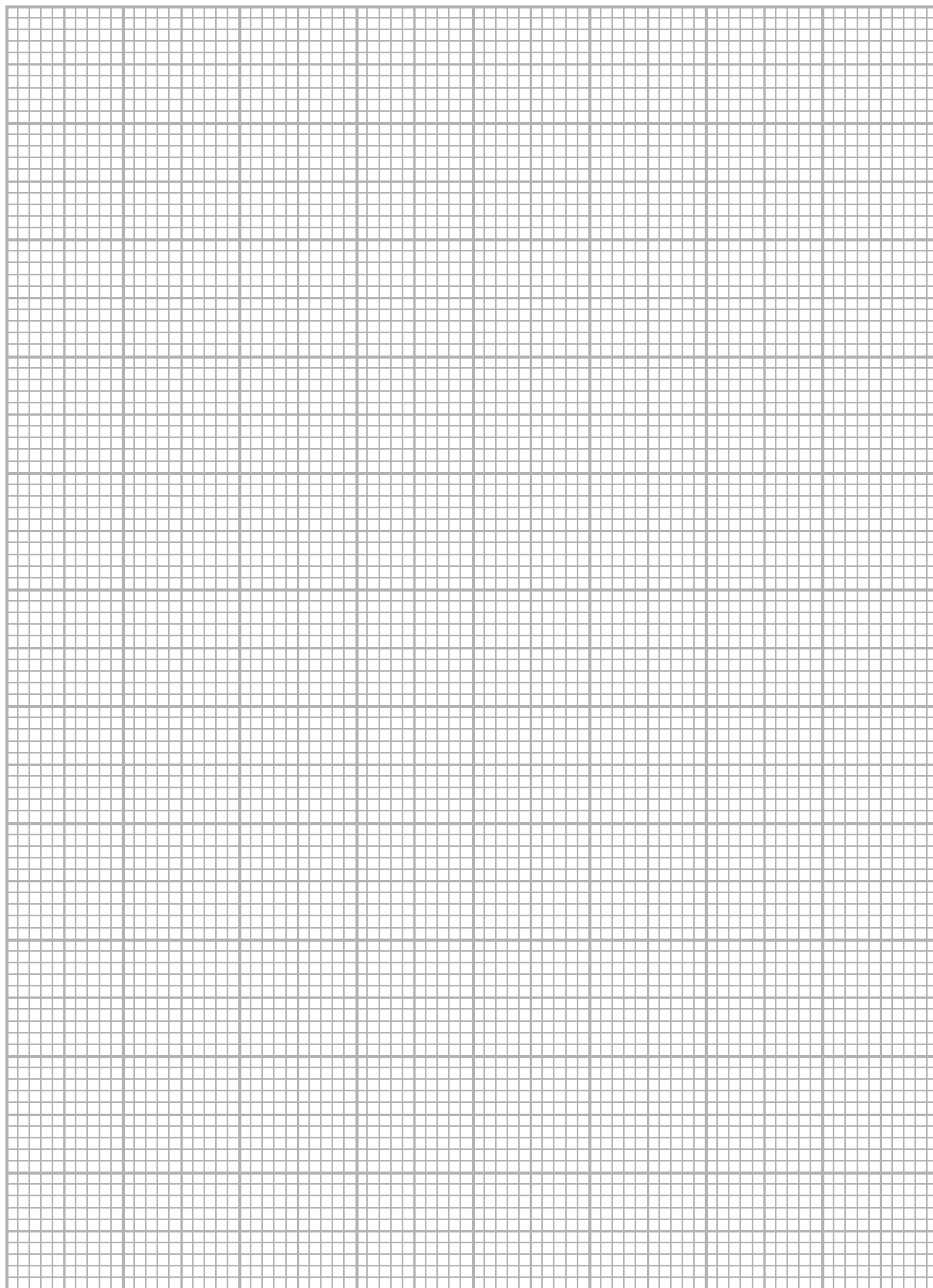
- (a) (i) Calculate the percentage change in mass when the carrot tissue was in a sucrose solution of 1.0 mol dm⁻³.

(2)

Answer..... %

- (ii) Plot a suitable graph to show the relationship between sucrose concentration and percentage change in mass. Join the points with straight lines.

(4)



- (iii) Using your graph, determine the concentration of sucrose solution which caused no change in mass of the carrot tissue.

(2)

- (iv) Explain why there is no change in mass of the carrot at this concentration.

(2)

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- (b) Five students investigated the effect of temperature on the movement of materials through the membranes of cells in a plant tissue. Beetroot was chosen for this investigation because the cells contain a red pigment called betalain.

One piece of beetroot was placed into a tube containing 15 cm^3 of water at 0°C and left for 15 minutes. The procedure was repeated for seven further temperatures between 10°C and 70°C .

Each piece of beetroot was removed and a sample of the fluid was placed in a colorimeter. The colorimeter was used to determine the intensity of the red colour of the fluid.

- (i) Name **two** variables, other than those mentioned above, which should be kept constant during this experiment.

(2)

(ii) State the dependent variable in this investigation.

(1)

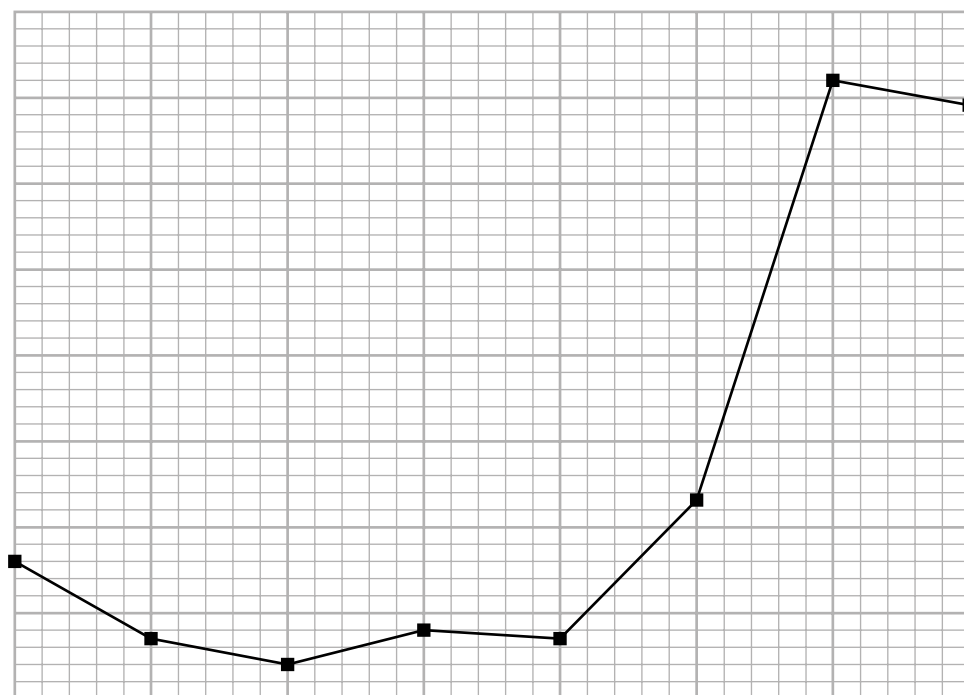
(iii) The table shows the results of this investigation.

Temperature /°C	Intensity of the red colour / a.u.						
	Student 1	Student 2	Student 3	Student 4	Student 5	Mean	Standard deviation
0	0.20	0.15	0.30	0.00	0.13	0.16	0.11
10	0.00	0.14	0.06	0.03	0.12	0.07	0.06
20	0.03	0.08	0.04	0.04	0.02	0.04	0.02
30	0.20	0.04	0.04	0.04	0.06	0.08	0.07
40	0.18	0.04	0.04	0.04	0.07	0.07	0.06
50	0.10	0.26	0.00	0.60	0.18	0.23	0.23
60	0.60	0.89	0.80	0.80	0.55	0.72	0.15
70	0.75	0.50	0.75	0.75	0.70	0.69	0.11

The graph showing the relationship between temperature and the mean intensity of red colour is incomplete.

Complete the graph.

(3)



(iv) Suggest why four of the readings were above zero a.u. at 0°C.

(2)

(v) The students concluded that as temperature increases, membrane permeability increases.

Using the information in the table and graph, criticise this conclusion.

(5)

(Total for Question 3 = 23 marks)

TOTAL FOR PAPER = 50 MARKS

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